GLOBAL SPATIAL VERIFICATION

Experiment 1 — Spatial Profile Evaluation Across 100 Trench Coordinates (N = 2,000)

MODELING METHOD

- Single PhysicsNeMo PINN surrogate
- Coordinate-by-coordinate spatial evaluation across 2,000 runs.
- Evaluates full 100-point spatial thickness profile simultaneously.

INPUT PARAMETERS

Dataset: 8000 process runs (800,000 points).

Temp: 150.0–300.0°C | TMA Mole Frac relative to total gas molecules: 0.0005–0.005.

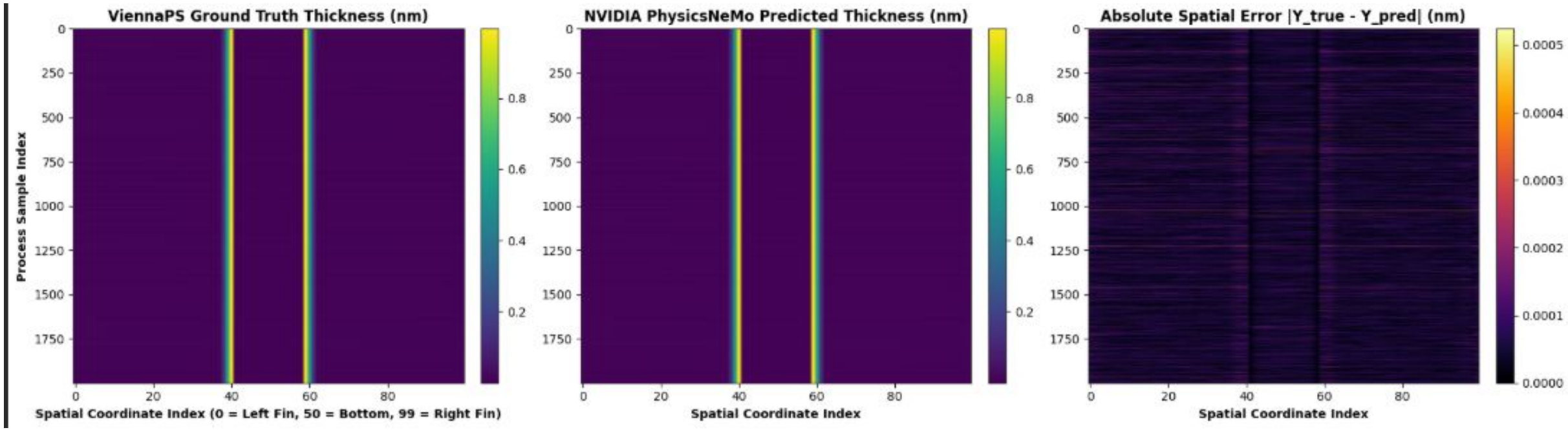
Pulse Time (0.05–2.0s), Cycle Count (4–20).

KEY METRICS

Global Spatial R^2: 0.999988

MAE: 0.0005016589 nm

Peak Error: 0.0005241185 nm


ViennaPS Ground Truth Thickness (nm)
NVIDIA PhysicsNeMo Predicted Thickness (nm)
Absolute Spatial Error |Y_true - Y_pred| (nm)
Process Sample Index
Spatial Coordinate Index (0 = Left Fin, 50 = Bottom, 99 = Right Fin)
Spatial Coordinate Index
Spatial Coordinate Index
0
250
500
750
1000
1250
1500
1750
0
20
40
60
80
0.2
0.4
0.6
0.8
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005

ERROR TAIL DIAGNOSTICS

Experiment 1 — Error Tail Isolation & Failure Mode Analysis (N = 2,000)

ERROR TAIL STATISTICS

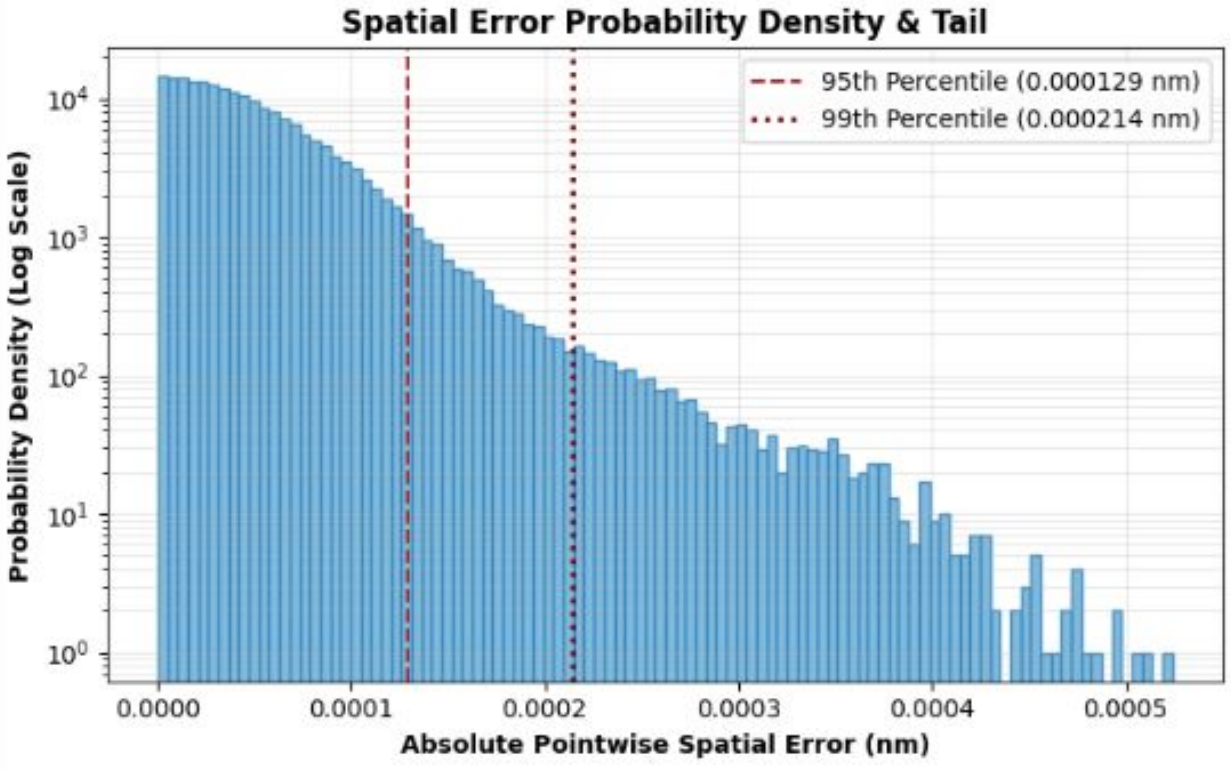
50th Percentile (Median): 0.00033 nm
95th Percentile Threshold: 0.00003953092 nm
99th Percentile Outlier: 0.0002144392 nm

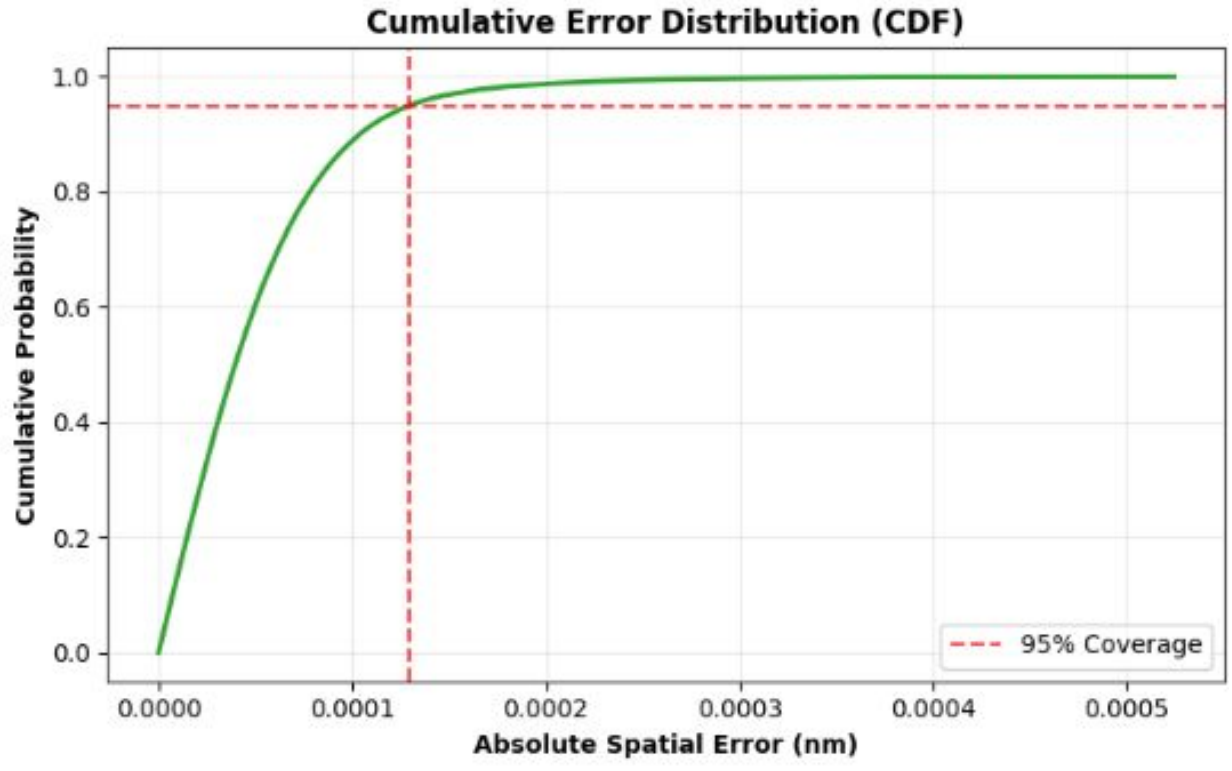
REGIONAL ERROR BREAKDOWN

Fin Peak MAE: 0.00005531573 nm
Sidewall MAE: 0.00005030218 nm
Trench Bottom MAE: 0.00004711389 nm

FAILURE DRIVERS

- Point 40 and 60 Discontinuity: Sharp 90° corner caused derivative smoothing.
- Nucleation Nonlinearity: Bad fits during transient initial growth cycles (N ≤ 5).


Spatial Error Probability Density & Tail
95th Percentile (0.000129 nm)
99th Percentile (0.000214 nm)
Probability Density (Log Scale)
10^4
10^3
10^2
10^1
10^0
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
Absolute Pointwise Spatial Error (nm)


Cumulative Error Distribution (CDF)
Cumulative Probability
1.0
0.8
0.6
0.4
0.2
0.0
95% Coverage
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
Absolute Spatial Error (nm)

PHYSICS LOSS UPGRADES

Model Optimization — Mathematical Formulations & Physics Interventions

SPATIALLY WEIGHTED DATA LOSS

$$L_{data} = \frac{1}{B} \sum_{i=1}^{B} \sum_{x=0}^{99} W_{spatial}(x) \cdot (\hat{Y}_{i,x} - Y_{i,x})^2$$

Bottom/Corners Weight: 3.0

NUCLEATION PHYSICS CONSTRAINT

$$L_{linear} = \frac{1}{B} \sum_{i=1}^{B} w(N_i) \cdot \frac{\partial^2 \hat{T}_{mean,i}}{\partial N^2}^2$$

Weight scales to 1.0 by N=10 cycles.

TOTAL OBJECTIVE FUNCTION

$$L_{total} = L_{data} + \lambda_{phys} L_{origin} + L_{kinetics} + L_{linear} + 0.1 L_{laplacian}$$

REGULARIZATION

- Weight Decay: 0.00005
- Validation Patience: 300 epochs
- Gradient Clipping: Max norm 0.1

4X4 FACTORIAL EXPERIMENT

Experiment 2 — Design Matrix, Setup & Execution Environment

Dataset size (right) Constraints (down)	20	100	1,000	8,000
P0 (no physics)	P0, n = 20	P0, n = 100	P0, n = 1000	P0, n = 8000
P1 (boundary)	P1, n = 20	P1, n = 100	P1, n = 1000	P1, n = 8000
P2 (kinetics)	P2, n = 20	P2, n = 100	P2, n = 1000	P2, n = 8000
P3 (multi-scale)	P3, n = 20	P3, n = 100	P3, n = 1000	P3, n = 8000

MATRIX DESIGN (16 MODELS)

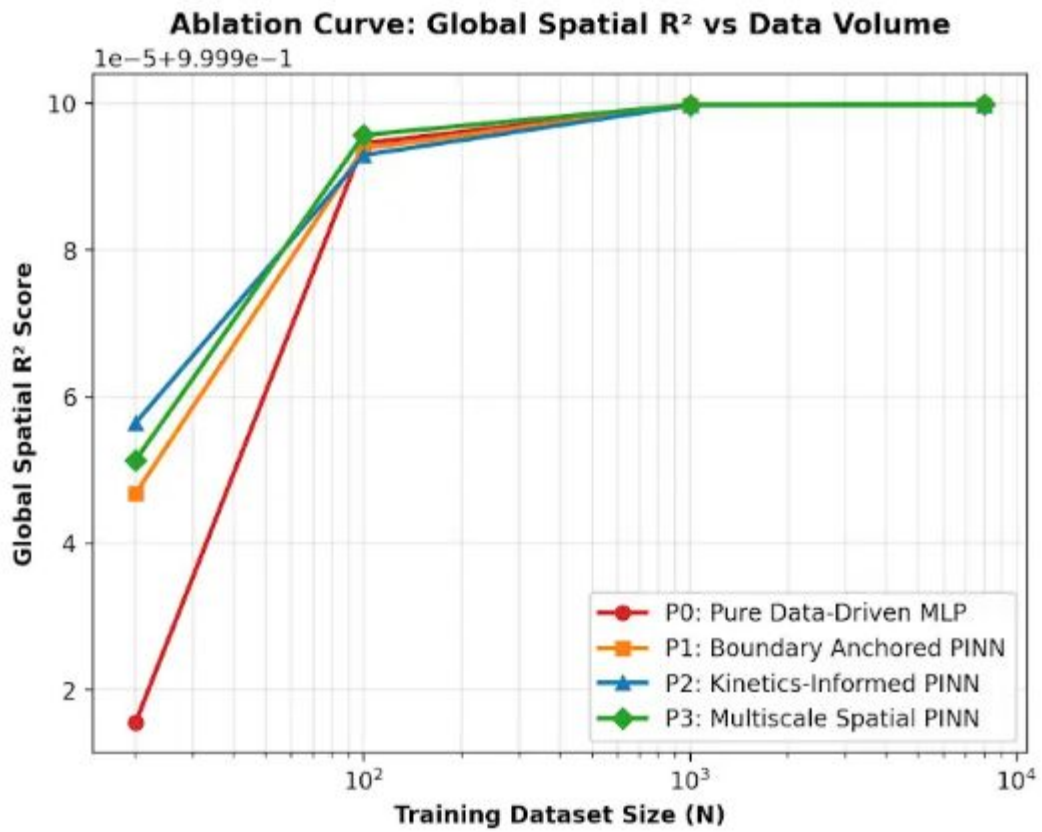
Data Tiers: 20, 100, 1,000, 8,000 samples.
P0: Black-box MLP | P1: Boundary Anchor.
P2: Kinetics-Informed | P3: Multiscale Spatial.

ENVIRONMENT SPEC

GPU: NVIDIA Tesla T4 (14 GB VRAM).
PyTorch 2.11.0 | Python 3.12.13.
Total Training Duration: 365s

LOW-DATA ABLATION RESULTS

Experiment 2 — Performance in Data-Scarce Regimes (N = 20 & N = 100)



RESULTS AT N = 20 SAMPLES

P0 MAE: 0.0010985 nm

P2 MAE: 0.0007294 (33.61% improvement)

P3 Tail Error: 0.0020577 nm

RESULTS AT N = 100 SAMPLES

P0 MAE: 0.0001584 nm

Stabilization: P3 smoothing stabilizes Conformality R^2 vs divergence in P0-P2.

HIGH-DATA & TRENCH ACCURACY

Experiment 2 — Peak Model Performance & Trench Bottom Resolution (N = 8,000)

PEAK MODEL PERFORMANCE (P3_N8000)

Global Spatial R^2: 0.9999988
Global MAE: 0.00004145
Overall Error Reduction: 12.47% vs unconstrained.
Step Conformality R^2: 0.958090

REGIONAL MAE GAINS TABLE

Trench Zone	P0 (Pure)	P3 (Multi)	Gain
Fin Peak	0.00005204	0.00004734	9.02%
Sidewall	0.00004721	0.00004195	11.14%
Bottom	0.00004502	0.00003676	18.36%
Corners	0.00005149	0.00004562	11.40%

EFFICIENCY & STORAGE AUDIT

Experiment 2 — Computational Footprint, Latency & Drive Storage Audit

EXECUTION RUNTIMES

Throughput: 1.0-2.4 million samples per second

Speedup: > 100,000x faster than TCAD.

Peak VRAM scaling: 19.52 → 543.18 mb

System RAM: 12.67 GB

DRIVE STORAGE FILE INVENTORY

Artifact File	Description	Footprint
model_P*.pt	16 Checkpoints	6.13 MB total
dataset.csv	Raw Dataset	22.84 MB
test_pred.npz	Binary Array	1.76 MB
system.json	Diagnostics	0.32 KB